Answers and talk notes

Math is Playful

What your student is learning

Tonight we solve a puzzle by starting SMALL — tracing simple cases until a pattern rule appears.

Answers

Ask your student to explain before you look.

1. Double it and add 1 is the rule. 7 doubled is 14, plus 1 is 15 moves for four discs.
2. A table lists discs 1, 2, 3 beside moves 1, 3, 7. That organizes the results.
3. With the rule, you can predict five discs: 15 × 2 + 1 = 31 moves.
4. The moves 1, 3, 7, 15, 31 follow a pattern.

Talk about it

Try your rule on two discs. Does 1 doubled give the real number of moves?

Home mission

Count out dry beans: a pile of 1, a pile of 3, and a pile of 7. Each pile is double the last plus one. Before counting, ask your student how many beans the next pile needs. Then count to test.

When they are stuck, ask

- What is the smallest version of this problem? Solve that first.
- Can you put your results in a table?
- What happens between one row and the next?

Try not to say

- Just try to solve the big one directly.
- Doubling is the whole rule.
- If it works once, the rule is right.